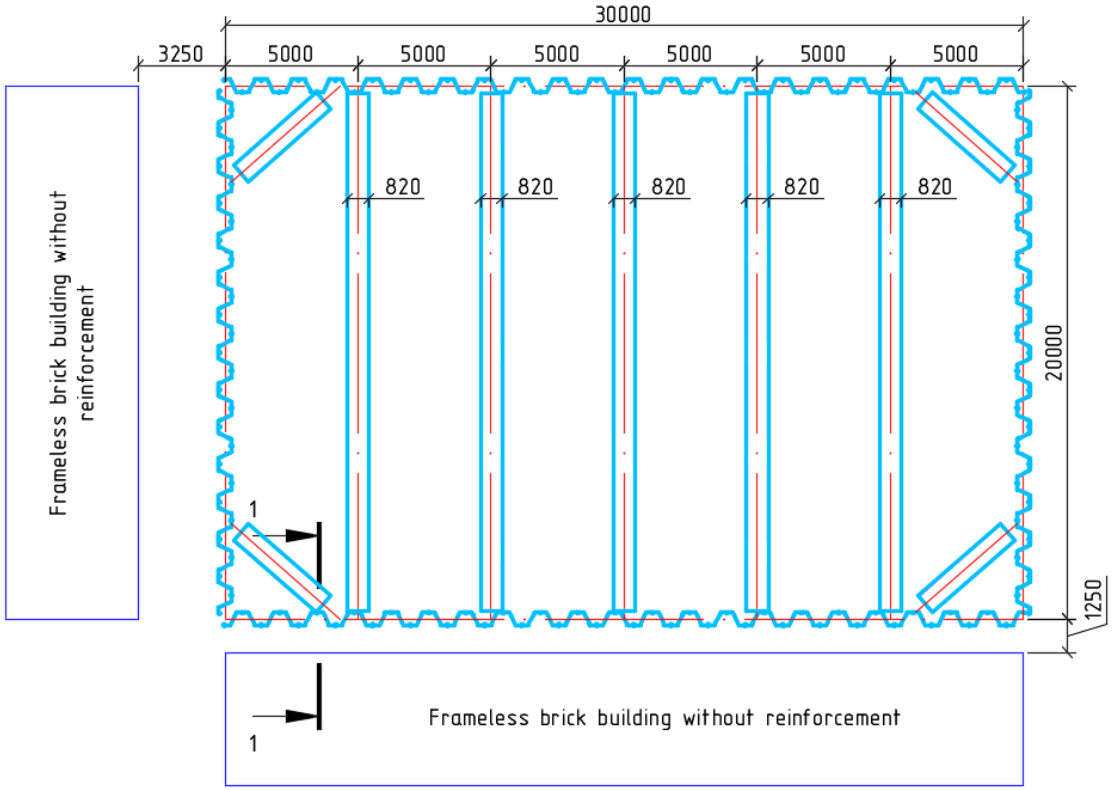
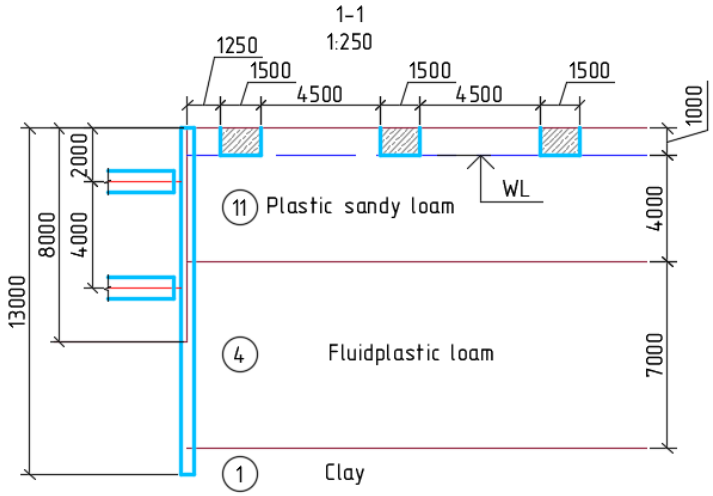
Initial data on ground conditions

Грунтовые условия												
Номер ИГЭ	Наименовани грунта	Мощность, м	Удельный вес, Кн/м3	Угол внутреннего трения, φ_{II}°	Удельное сцепление,cII,кПа	Удельный вес твердых частиц грунта, γ_s, кН/м^3	Влажность, W	Влажность на границе текучести, WL	Влажность на границе пластичности, WP	Коэффициент фильтрации, Кф, см/с	Модуль деформации, E, МПа	Уровень грунтовых вод, HWL, м
ИГЭ №11	Супесь	5	18,6	20	7	26,3	0,18	0,21	0,16	2,5·10-5	16	1
ИГЭ №4	Суглинок	7	18,6	15	14	26,5	0,23	0,24	11	2,3·10-7	4	
ИГЭ №1	Глина	-	20	20	30	26,9	0,25	0,46	0,27	3,1·10-8	20	

	Показатель текучести, IL	Удельный вес сухого, γ_d	Коэффициент пористости, e	w_{sat}	Удельный вес в водонасыщенном, γ_{sat}	Угол внутреннего трения, φ_{II}°	Удельное сцепление,cI,кПа	Коэффициент пуассона, μ
ИГЭ №11	0,4	15,76	0,668	0,25	19,769	18,182	6,3636	0,3
ИГЭ №4	1	15,12	0,752	0,28	19,416	13,636	12,727	0,35
ИГЭ №1	-0,1	16	0,681	0,25	20,052	18,182	27,273	0,42

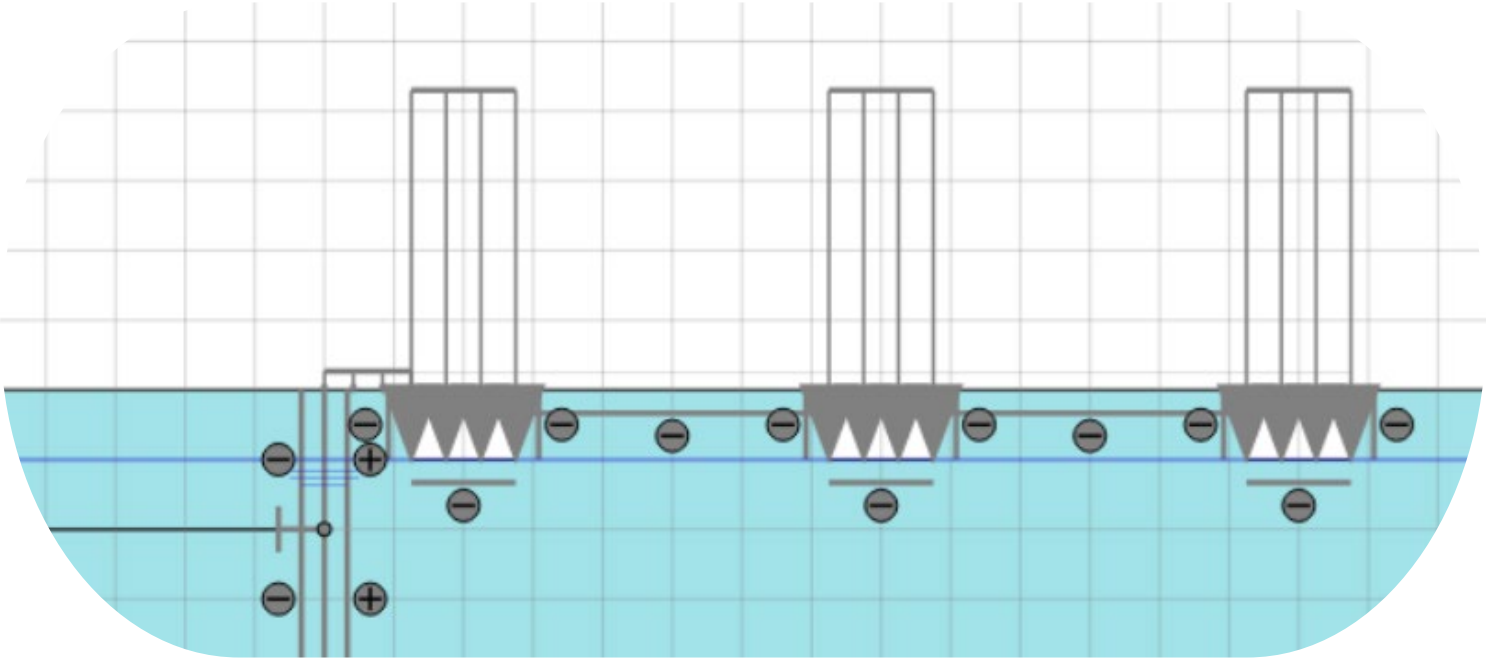

30000
3250
5000
5000
5000
5000
5000
5000
820
820
820
820
820
20000
1250
1
Frameless brick building without reinforcement
Frameless brick building without reinforcement
1
1


1-1
1:250
1250
1500
4500
1500
4500
1500
1000
2000
8000
4000
13000
WL
11 Plastic sandy loam
4000
4 Fluidplastic loam
7000
1 Clay

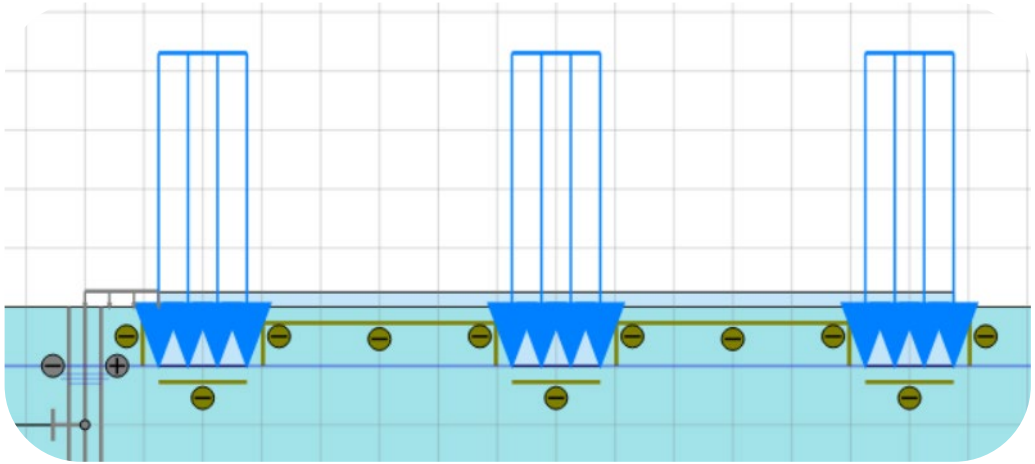
Model in PLAXIS 2D



Calculation phases, initial phase



Calculation phases, I phase

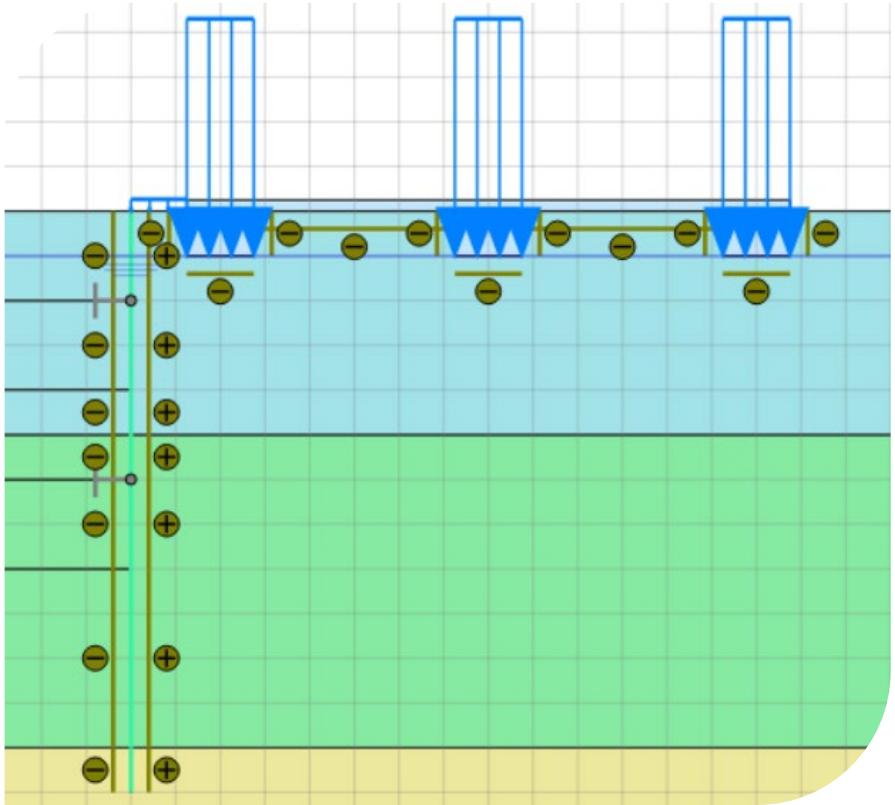


- Setting an existing building;
- Foundation sedimentation;

II phase → Zeroing out deformations

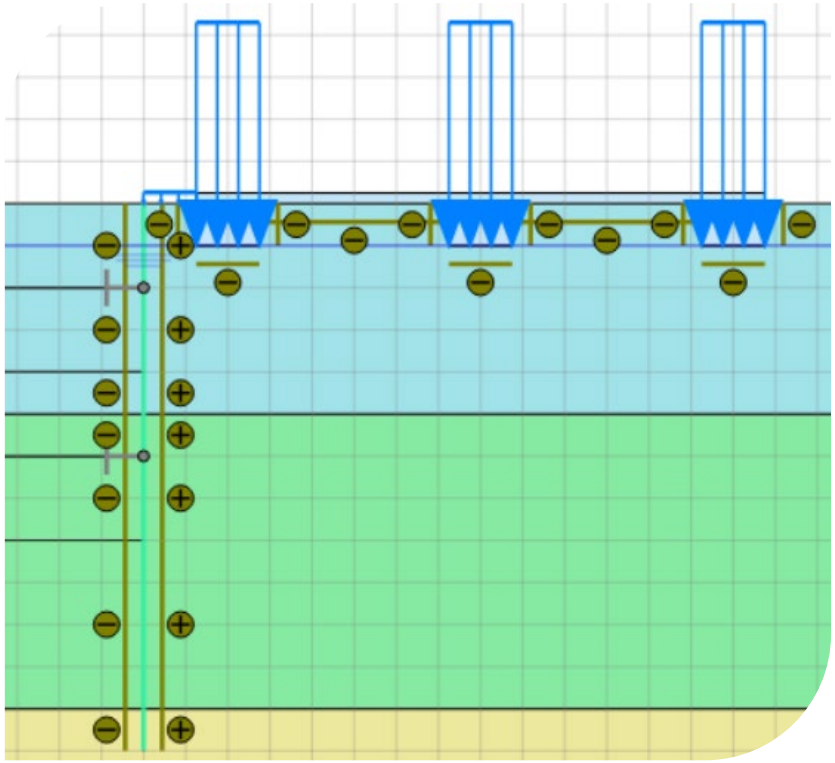
Reset displacements to zero	☑

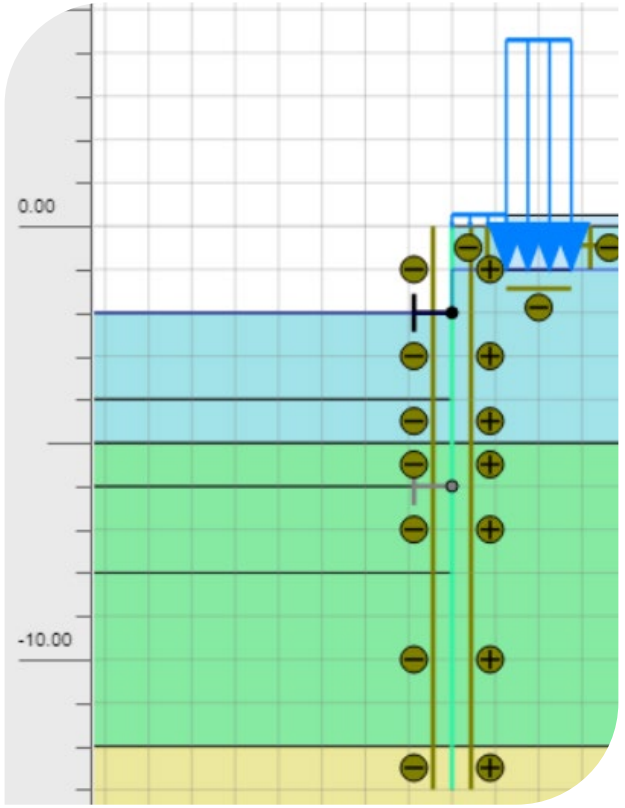
Calculation phases, III phase



- Sinking of the sheet pile wall;
- Setting a uniformly distributed load on the edge;

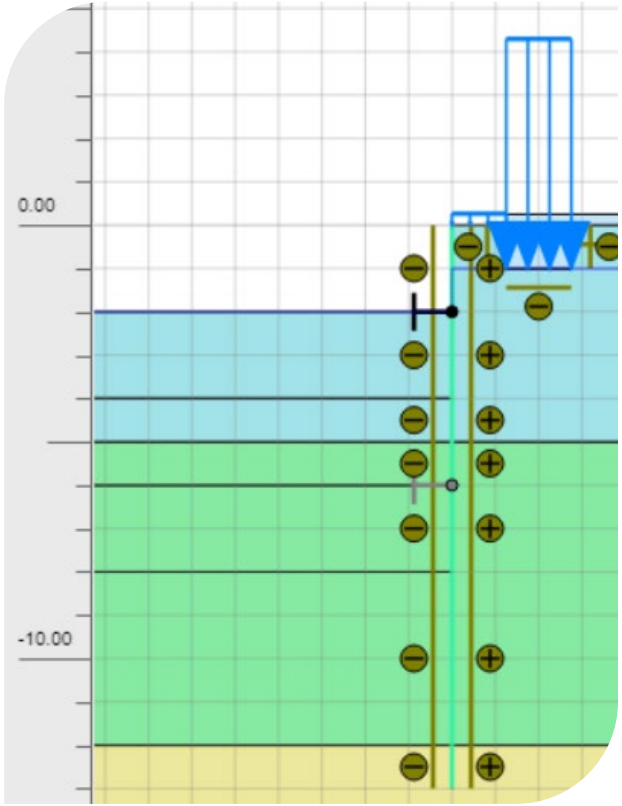
Calculation phases, IV phase

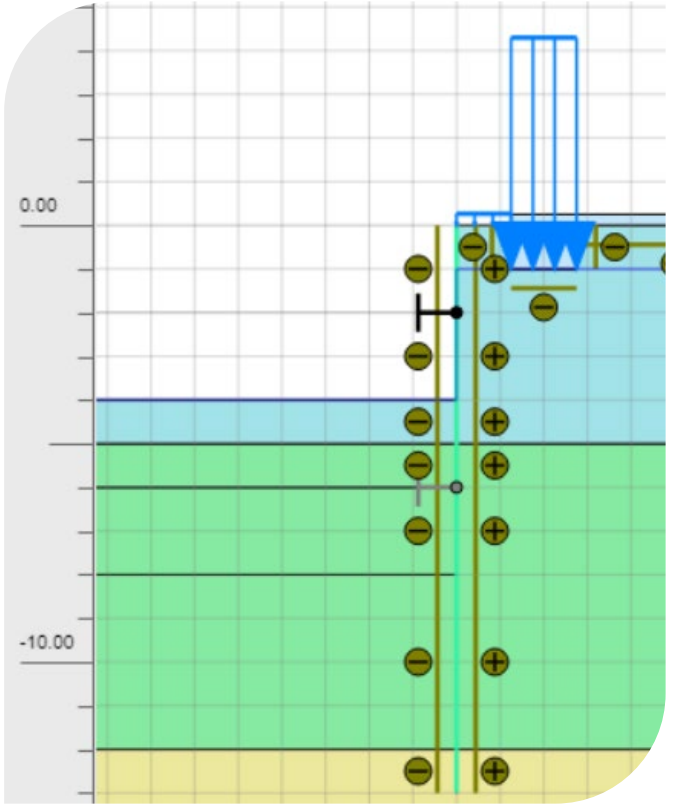




- Excavation of a layer with a depth of 2 m;
- Installation of the first level of struts;

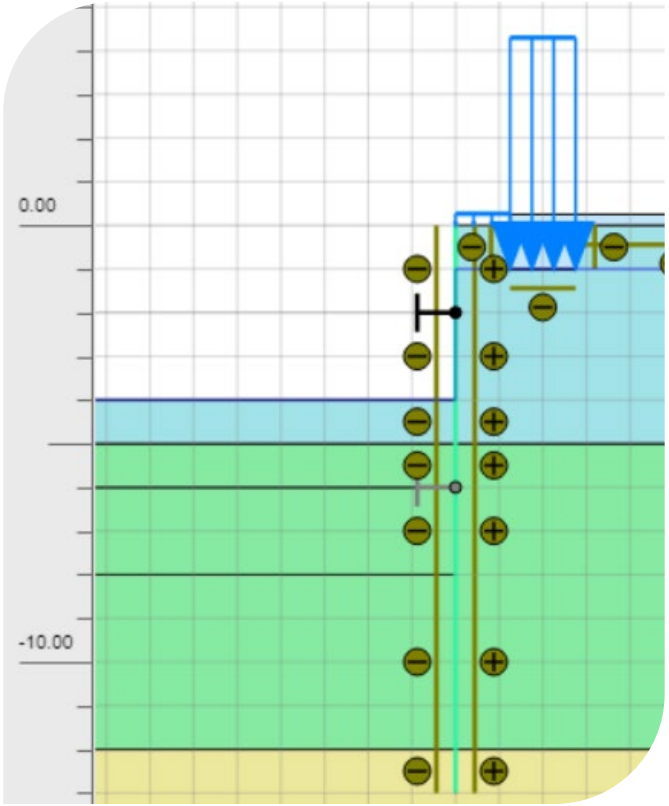
Calculation phases, V phase

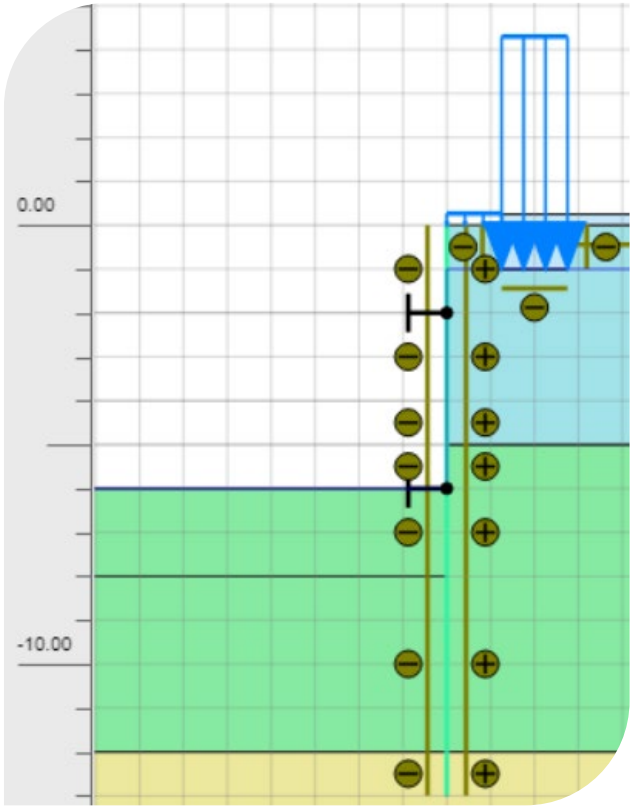




Excavation of a layer with a depth of 2m

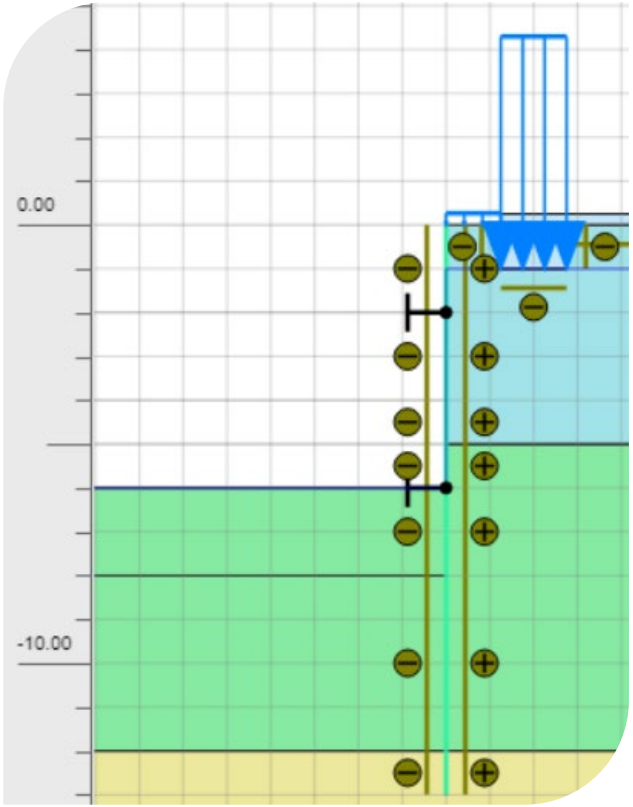
Calculation phases, VI phase

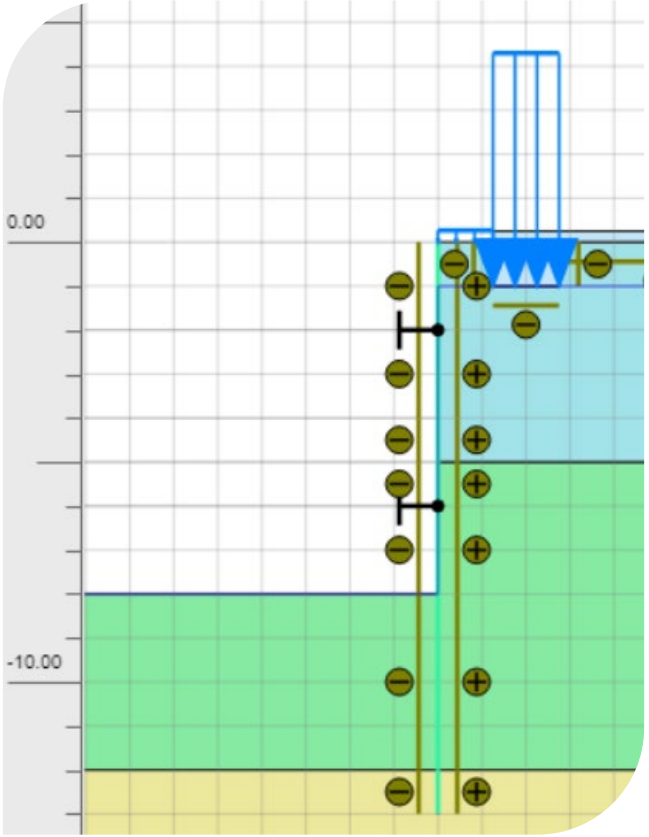




- Excavation of a layer with a depth of 2 m;
- Installation of the second level of struts;

Calculation phases, VII phase





Excavation of a layer with a depth of 2m

Selection of the tongue-and-groove fencing of the excavation and the section of the struts

The selection was carried out through several iterations to achieve the highest material consumption and meet the conditions for precipitation of neighboring buildings according to the annex to the Joint Venture 22.13330.2016

Maximum draft for a multi-storey frameless building of the II-th category of technical condition: *3,0 cm*

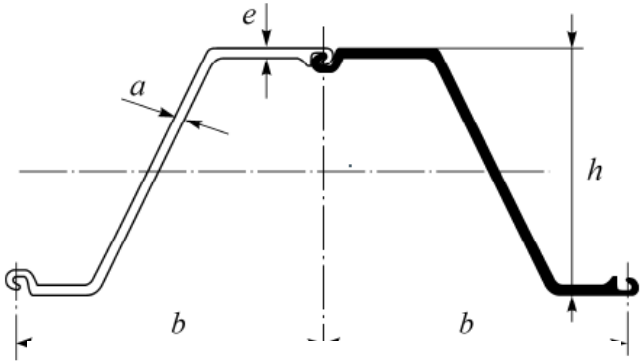
The maximum relative difference in the sediment of free-standing foundations: *0,0010*

The section of the spacer structure was selected according to GOST 10704-91 "Straight-welded steel pipes"

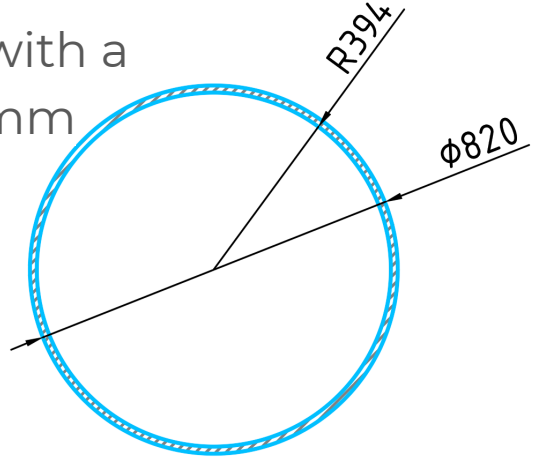
Selected sections

- Sheet piling: Larsen sheet piling AZ - 38

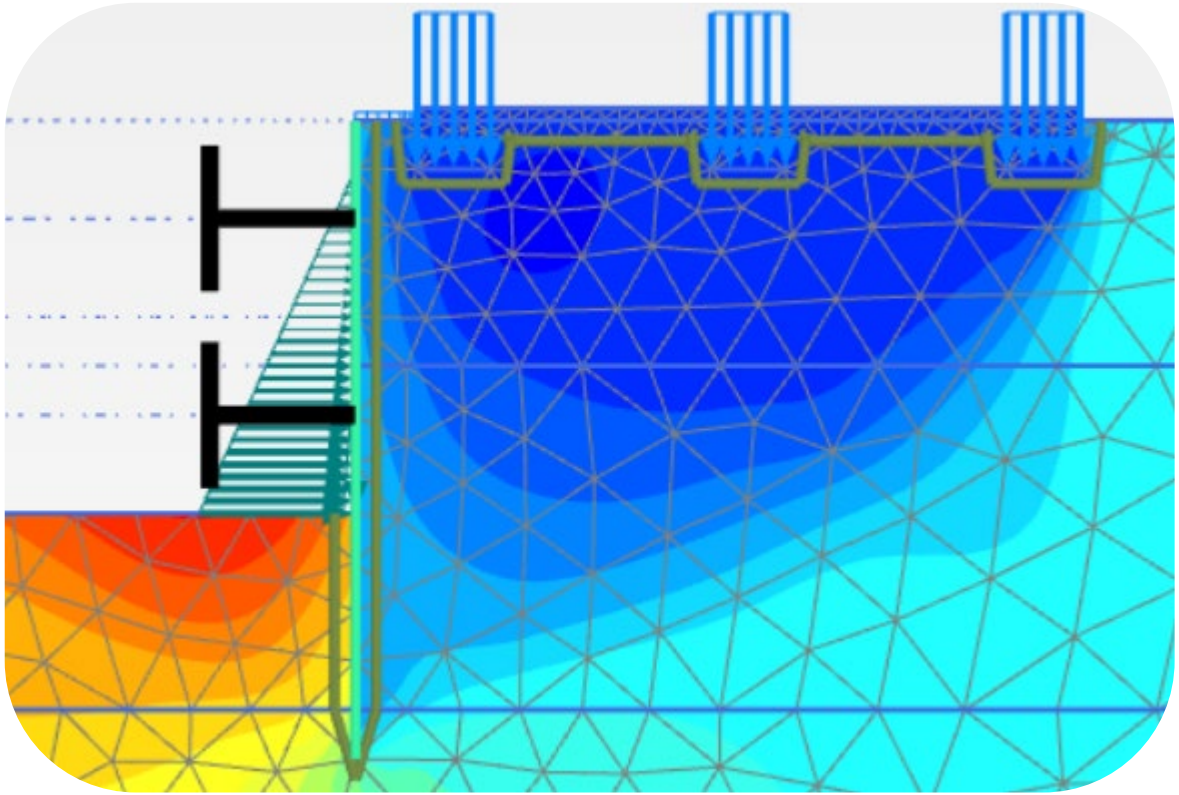
b = 630 мм
h = 461 мм
e = 19,0 мм
a = 15,0 мм



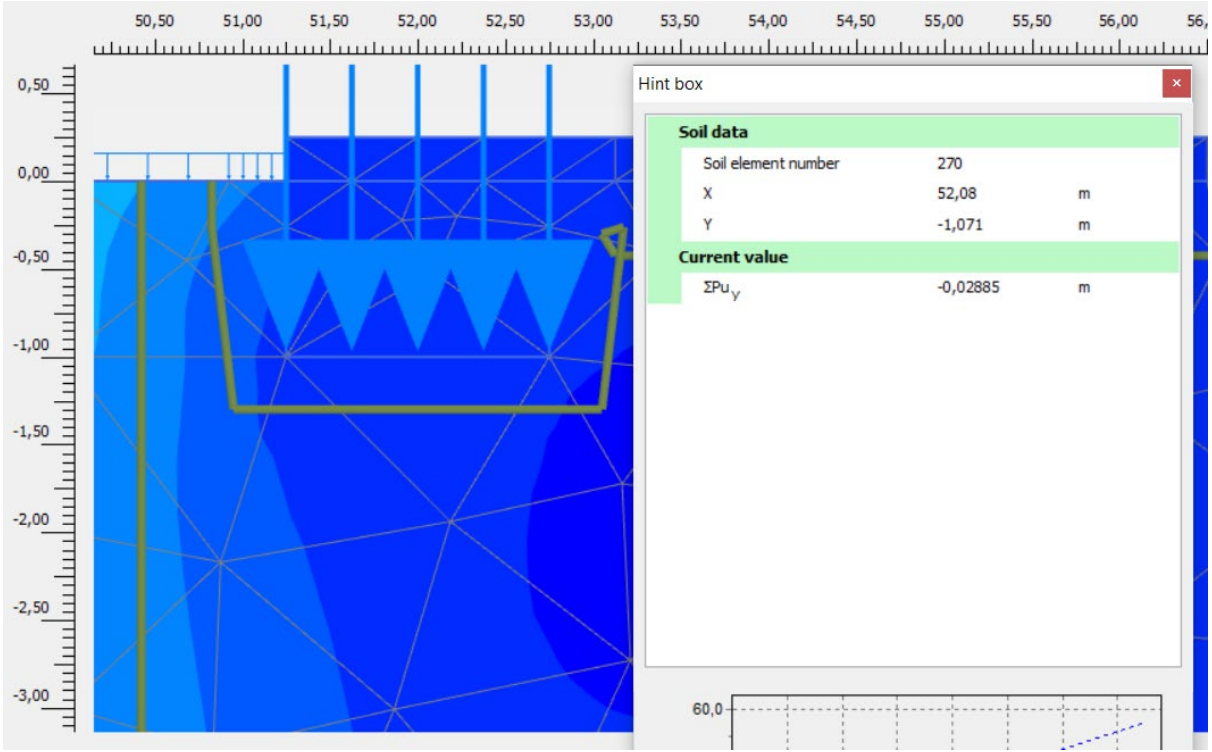
- Struts: Electric welded straight-seam steel pipe with a diameter of 820 mm with a wall thickness of 16 mm



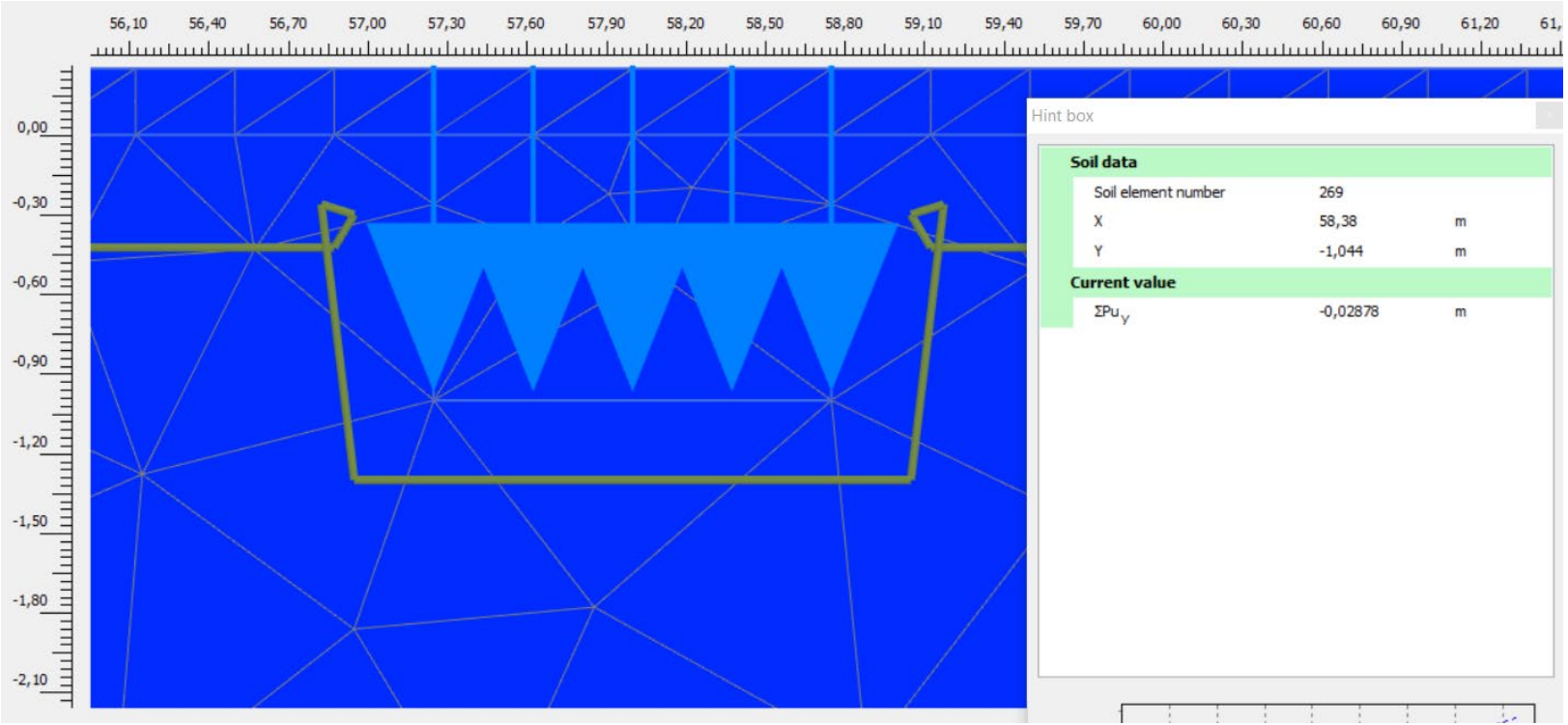
Precipitation of foundations of nearby buildings



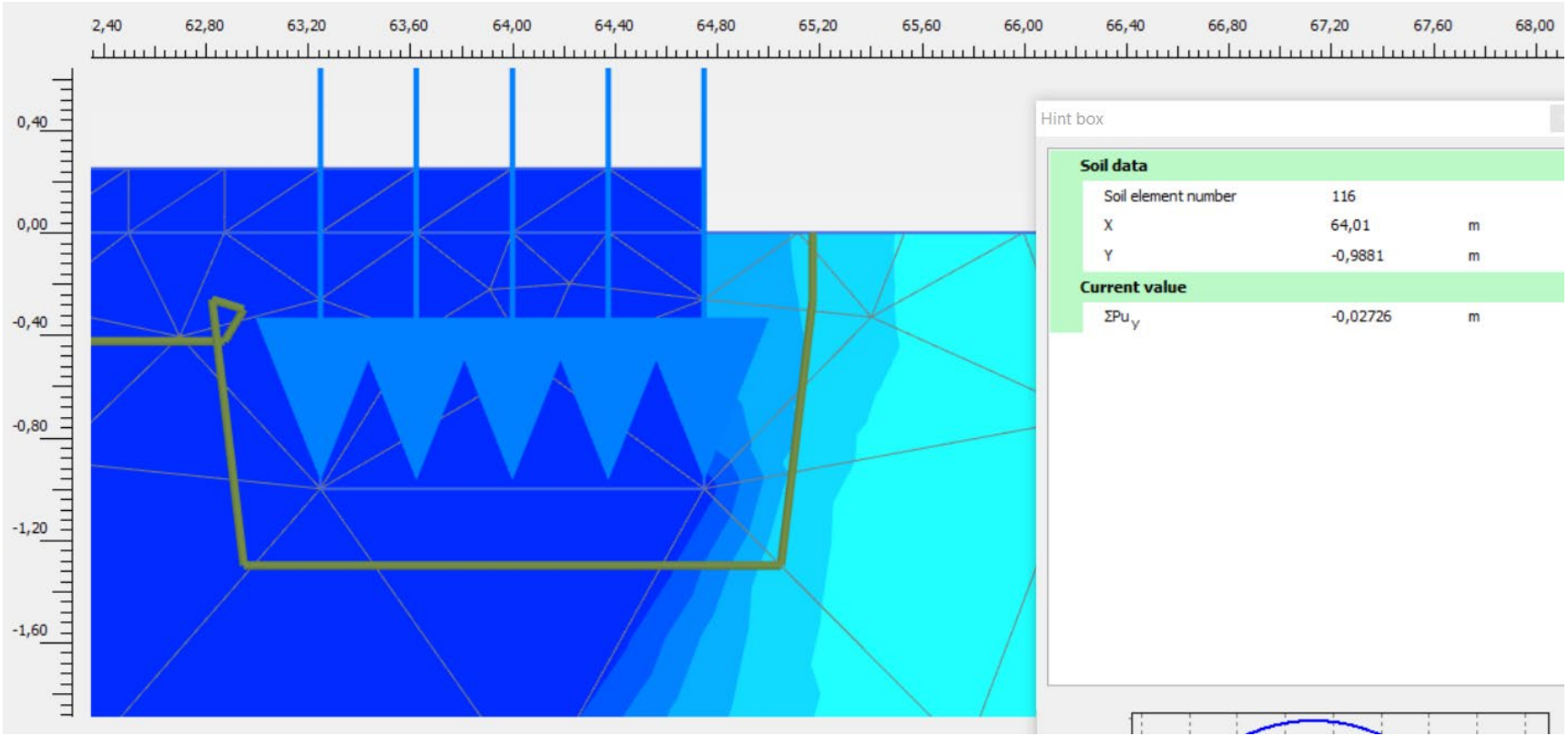
First foundation



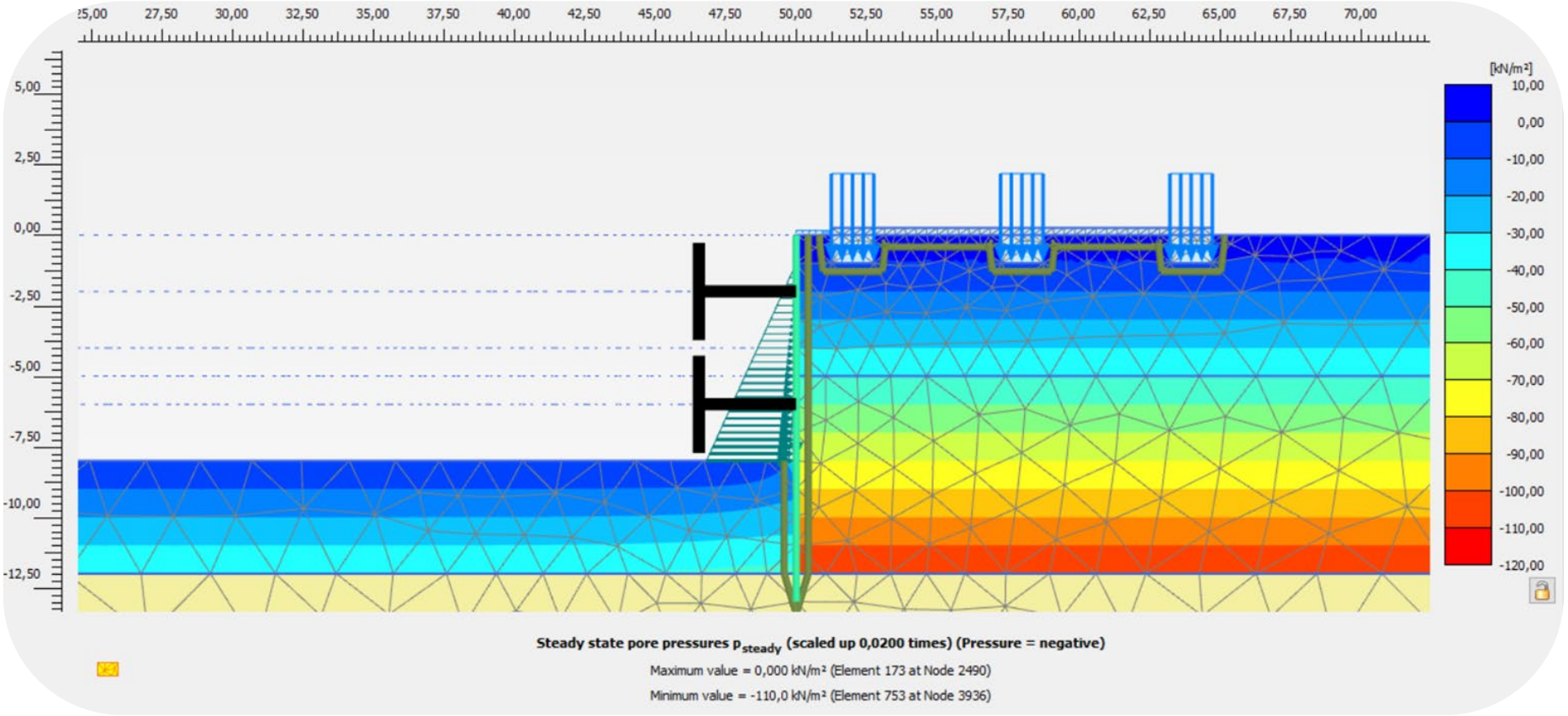
Second foundation



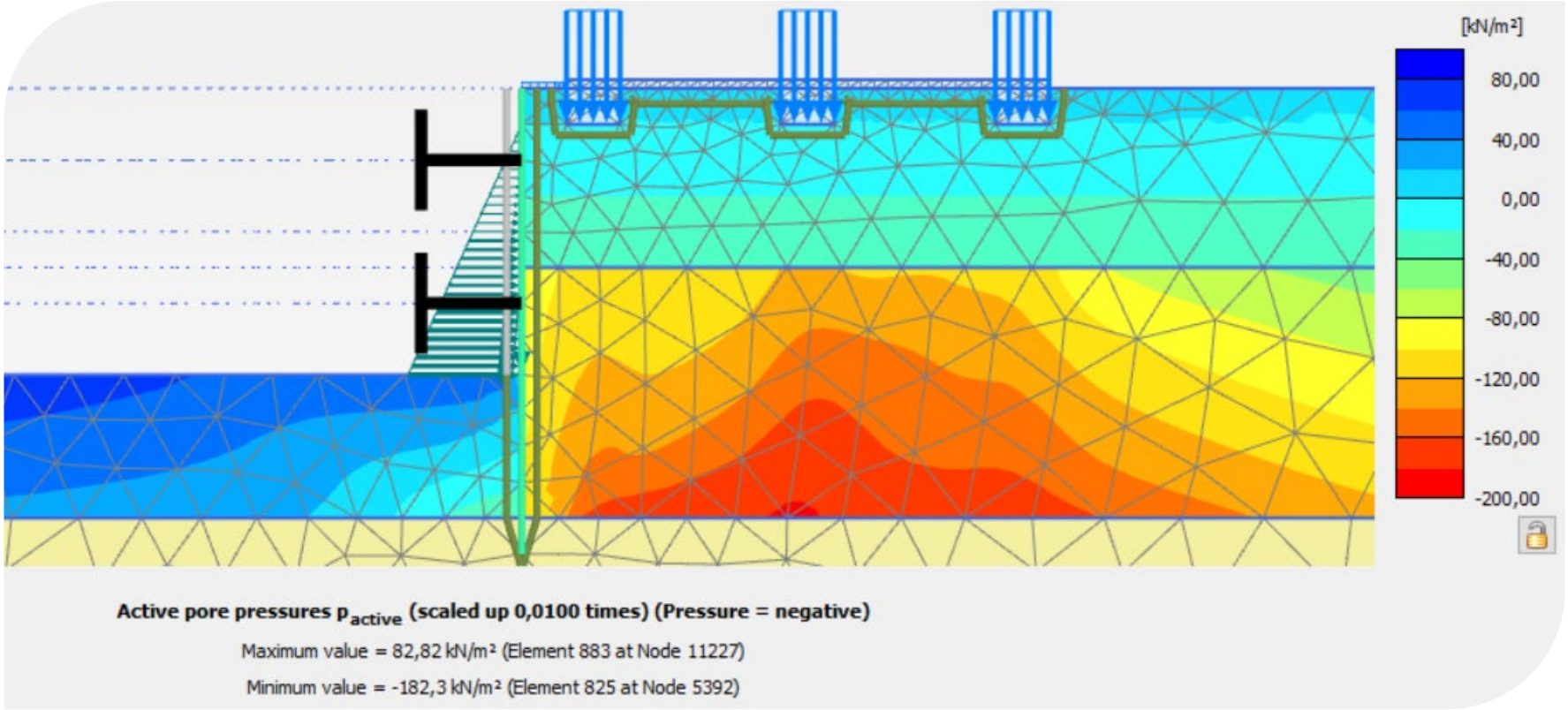
Third foundation



Steady state pore pressures, phase VII



Active pore pressure, phase VII

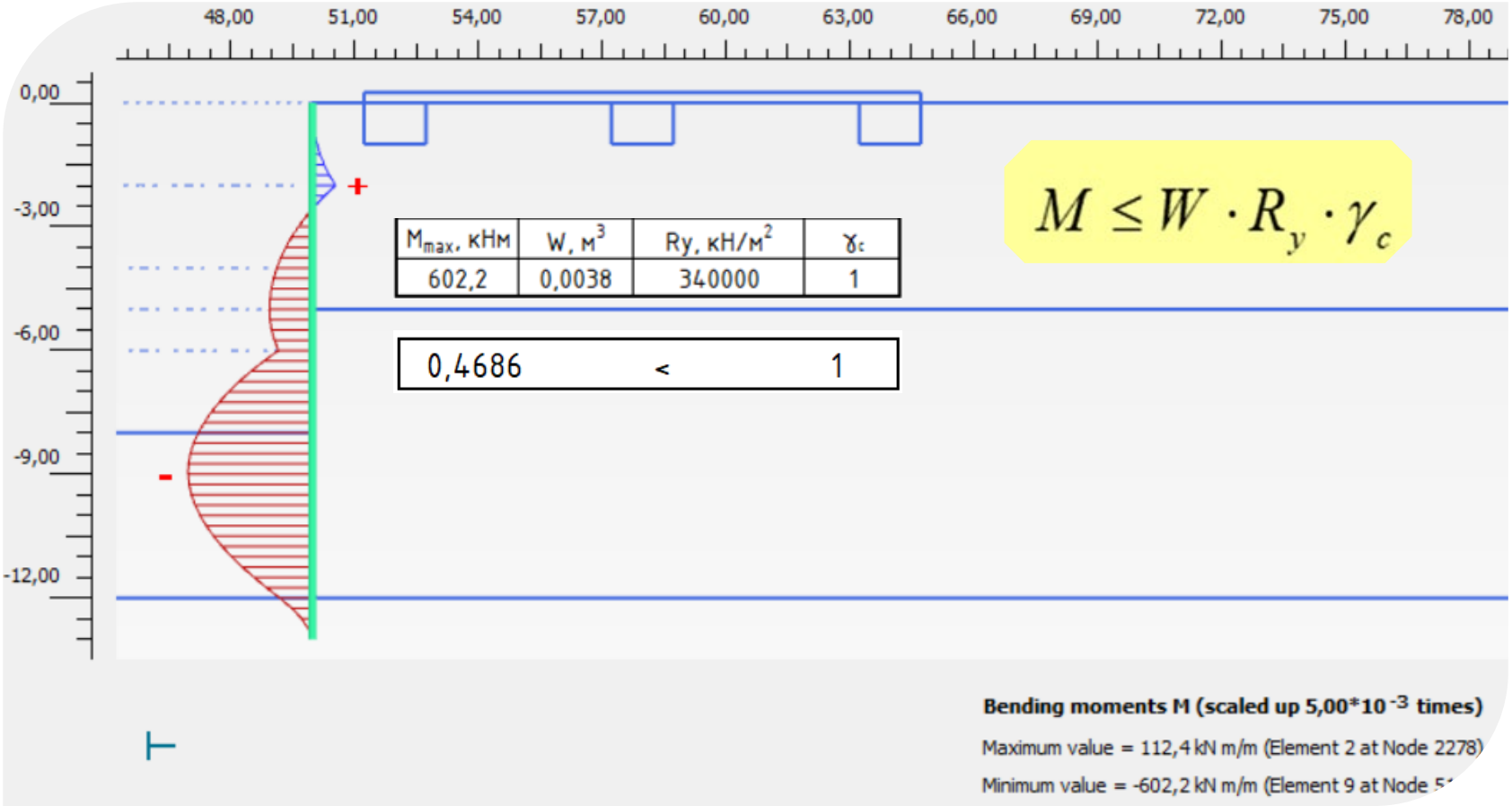


Active pore pressures p_{active} (scaled up 0,0100 times) (Pressure = negative)

Maximum value = 82,82 kN/m² (Element 883 at Node 11227)

Minimum value = -182,3 kN/m² (Element 825 at Node 5392)

Calculation of sheet pile fencing for bending, phase VII



M_{max}, кНм	W, м^3	Ry, кН/м^2	γ_c
602,2	0,0038	340000	1

0,4686 < 1